

Q-1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

ANSWER:-

1. GPU Identification

GPU Brand	GPU Model	Type
Intel	Intel UHD Graphics 620	Integrated GPU

Explanation: -

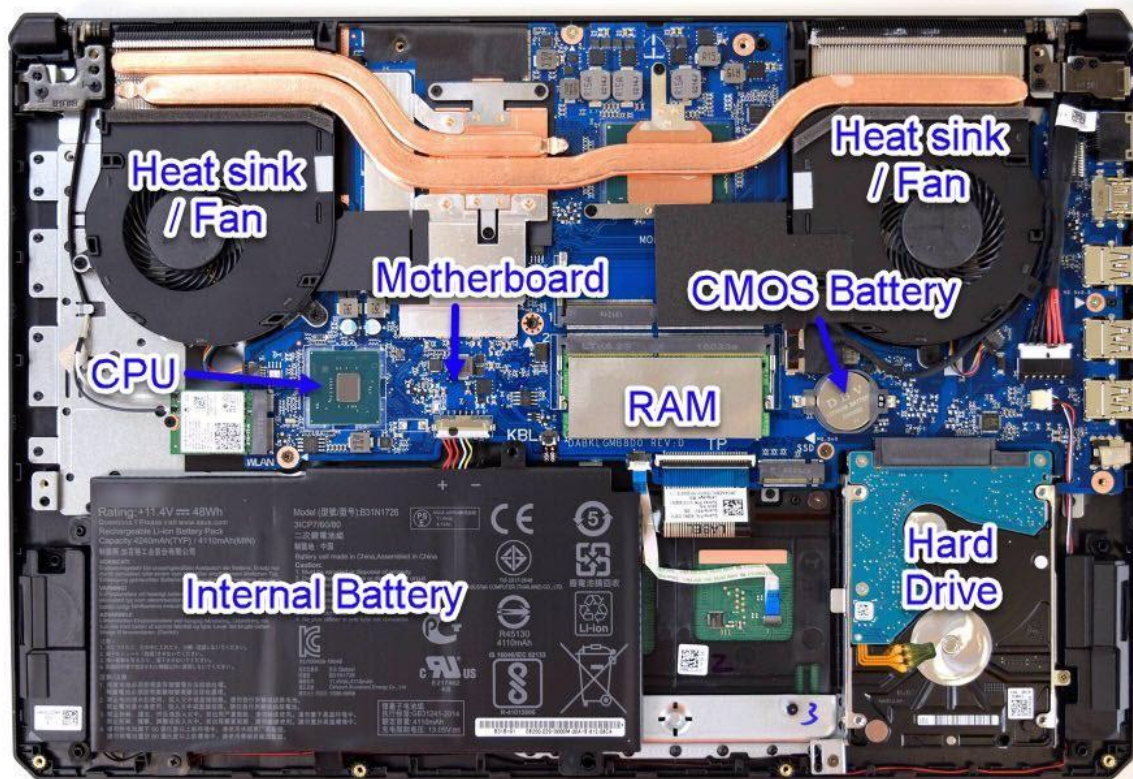
The Dell Latitude 5500 uses **Intel UHD Graphics 620**, which is built into the Intel processor. It is an **integrated GPU**, meaning it shares system RAM and does not have separate dedicated video memory.

How to check on your laptop:

1. Press **Windows + X** → Select **Device Manager**
2. Open **Display adapters**
3. Check the GPU model listed there.

Q-2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

ANSWER:-



Q-3 Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

ANSWER:-

Factor	Air Cooling	Liquid Cooling
Cost	Low cost; uses a fan and heatsink, making it affordable.	High cost; requires a pump, radiator, tubes, and coolant.
Cooling Efficiency	Good cooling performance for normal gaming and daily use.	Higher cooling efficiency; removes heat faster and supports high-performance GPUs.
Maintenance	Easy maintenance; only requires cleaning dust from fans and heatsinks.	More maintenance; requires checking coolant, pump, and possible leaks.
Noise Level	Fans can become noisy at high speeds.	Usually quieter because the radiator and pump handle heat more efficiently.
Installation	Simple installation and fewer components.	More complex installation and requires more space inside the PC case.

Q-4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

ANSWER:-

I would choose liquid cooling for the GPU because a gaming PC used for graphics-heavy games and IPL streaming needs better heat control during long sessions. Liquid cooling provides higher cooling efficiency and keeps GPU temperatures lower compared to air cooling. It also reduces fan noise, which is useful while streaming. Although it is more expensive and requires more maintenance, liquid cooling gives better performance for a high-end gaming and streaming setup.